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Socio-Environmental and Academic Determinants of Student Suspension Rates: A Census-Level Analysis

Data

The empirical basis for this study is a multidimensional dataset constructed by consolidating various public-sector indicators characterizing the educational and socio-economic landscape of the 2022-2023 academic year. We primarily focused on census tract-level metrics to ensure a granular understanding of neighborhood contexts. The educational data was derived from state and local performance records, specifically capturing secondary school proficiency in Mathematics and English Language Arts, as well as rates of school engagement through chronic absenteeism metrics. To account for the broader economic environment, we integrated household-level data regarding participation in the Supplemental Nutrition Assistance Program (SNAP) and student-level eligibility for Free and Reduced Price Lunch (FRPL) programs. Finally, to explore the potential impact of physical neighborhood characteristics, we included tree canopy and green coverage data from 2022. This synthesis of academic, economic, and environmental indicators allows for a comprehensive examination of any factors that correlate with student disciplinary outcomes across the region.

Variables

The following table provides a summary of the variables utilized in this analysis, including their specific metric and theoretical significance within the study of student suspension rates.

Variable	Description	Role
Student Suspension Rate	Percentage of students receiving one or more out-of-school suspensions in 2023.	Dependent
Math High Proficiency	Percentage of students achieving 'High' proficiency on 2023 math standardized tests.	Independent (Academic)
English High Proficiency	Percentage of students achieving 'High' proficiency on 2023 ELA standardized tests.	Independent (Academic)
Chronic Absenteeism	Percentage of students missing 10% or more of school days.	Independent (Behavioral)
SNAP Participation	Percentage of households receiving SNAP benefits; a proxy for neighborhood poverty.	Independent (Economic)
FRPL Eligibility	Percentage of students eligible for Free/Reduced Price Lunch; a proxy for family-level disadvantage.	Independent (Economic)
Green Coverage	Percentage of land area with tree canopy or vegetation as of 2022.	Independent (Environmental)

Each variable represents a critical dimension of the student's neighborhood and school experience, allowing for a multifaceted assessment of disciplinary predictors within the subsequent regression models.

Statistical Analysis

The statistical framework for this study involved a multi-stage data integration and analytical workflow designed to maximize the reliability of census-level estimates. The data merging process was executed in R using a systematic full-join approach based on unique Census Tract Identifiers (geoid). To ensure the integrity of the linear models, a rigorous missingness threshold was applied, where any census tract containing more than ninety percent missing data for the key indicators was excluded from the final analytical sample.

Our primary analytical strategy employed a hierarchical Multiple Linear Regression (MLR). This methodology allowed us to compare a baseline model focused solely on academic core proficiencies (Model 0) against a final integrated model (Model 1) that incorporated the broader socio-behavioral and environmental predictors. By analyzing the change in explanatory power (R^2) between these models, we could isolate the relative importance of neighborhood-level factors—such as economic disadvantage and chronic absenteeism—above and beyond purely academic performance. Standard diagnostic checks for collinearity, normality of residuals, and homoscedasticity were performed to assess the robustness of the linear estimates.

Results

The hierarchical regression analysis revealed significant insights into the determinants of student suspension rates, with the integrated model provide a statistically superior fit compared to the academic baseline.

Model Performance and Comparison

Model	R	R^2	Adjusted R^2	RMSE	R^2 Change	df1	df2	p
M_0	0.187	0.035	0.029	3.033	0.035	2	312	.004
M_1	0.321	0.103	0.086	2.943	0.068	4	308	< .001

The addition of socio-environmental predictors in Model 1 resulted in a significant R^2 change of 0.068 ($p < .001$), indicating that factors like chronic absenteeism and economic disadvantage explain an additional 6.8% of the variance in suspension rates that academic proficiency alone cannot capture.

Analysis of Variances (ANOVA)

Model	Source	Sum of Squares	df	Mean Square	F	p
M ₀	Regression	104.563	2	52.281	5.683	.004
	Residual	2870.486	312	9.200		
M ₁	Regression	307.298	6	51.216	5.913	< .001
	Residual	2667.751	308	8.662		

Analysis of Coefficients

The final model coefficients identify the primary drivers of student discipline while providing specific "options" for student-led or institutional intervention.

Variable	<i>B (Unstd.)</i>	<i>Std.Error</i>	<i>β (Std.)</i>	<i>t</i>	<i>p</i>
(Intercept)	4.015	1.241		3.235	.001
English Proficiency	0.017	0.017	0.117	0.995	.321
Math Proficiency	-0.049	0.020	-0.313*	-2.429	.016
SNAP %	0.018	0.022	0.055	0.812	.418
Chronic Absenteeism	0.050	0.015	0.247*	3.273	.001
FRPL %	-0.040	0.010	-0.331*	-3.955	< .001
Green Coverage	-0.011	0.023	-0.030	-0.467	.641

*Significant at $p < .05$

The Story: Interpreting Options for Success

The coefficients from Model 1 provide actionable insights into the "story" of school discipline. When we translate these statistical values into meaningful real-world shifts—such as a 10 percentage point improvement in a tract's educational profile—the "story" of the data becomes clear and actionable.

First, consider **Academic Achievement**. For every **10 percentage point increase** in students achieving high math proficiency within a census tract (e.g., moving from 20% to 30%), the predicted suspension rate is expected to decrease by **0.49 percentage points** ($B = -0.049$; $-0.049 \times 10 = -0.49$). This result illustrates that academic mastery acts as a protective shield; as students' competence in math grows, their disciplinary risk falls. This highlights a clear option for schools: investing in mathematics tutoring and core academic support is not just an educational strategy—it is a disciplinary one that yields measurable improvements in school climate.

Second, the data points to **Engagement and Attendance**. For every **10 percentage point increase** in chronic absenteeism, the suspension rate is predicted to increase by **0.50 percentage points** ($B = 0.050$; $0.050 \times 10 = 0.50$). This link is nearly identical in magnitude to the academic effect, but in the opposite direction. It suggests that the simple act of being present in school significantly alters a tract's disciplinary profile. This presents a second option: behavioral interventions focused solely on reducing absenteeism (connecting students back to the classroom) could be as effective as academic interventions in lowering suspension rates.

Finally, the relationship with **Economic Disadvantage (FRPL)** shows that for every **10 percentage point increase** in FRPL eligibility, the suspension rate actually decreases by **0.40 percentage points** in this complex multivariate model ($B = -0.040$; $-0.040 \times 10 = -0.40$). While this negative relationship appears counterintuitive, it underscores the finding that poverty does not have a single, direct path to discipline; its effects are often mediated through the very academic and attendance factors we have highlighted. For students, this means that even in the most economically challenged neighborhoods, boosting math proficiency and improving attendance remain viable, powerful options for disrupting the cycle of school suspension.

Critical Assessment of Assumptions

A rigorous examination of the diagnostic plots reveals that the core assumptions of Ordinary Least Squares (OLS) regression are notably violated. These violations serve as a critical warning that our results, while substantively illustrative, are based on a model that does not fully fit the underlying data structure.

Figure 1: Residuals vs. Predicted Values (Heteroscedasticity)

Residual	-7.325	23.664	8.659e-16	2.915	315
Std. Predicted Value	-2.647	4.833	-4.013e-17	1.000	315
Std. Residual	-2.720	8.253	-5.154e-4	1.007	315

Residuals vs. Predicted

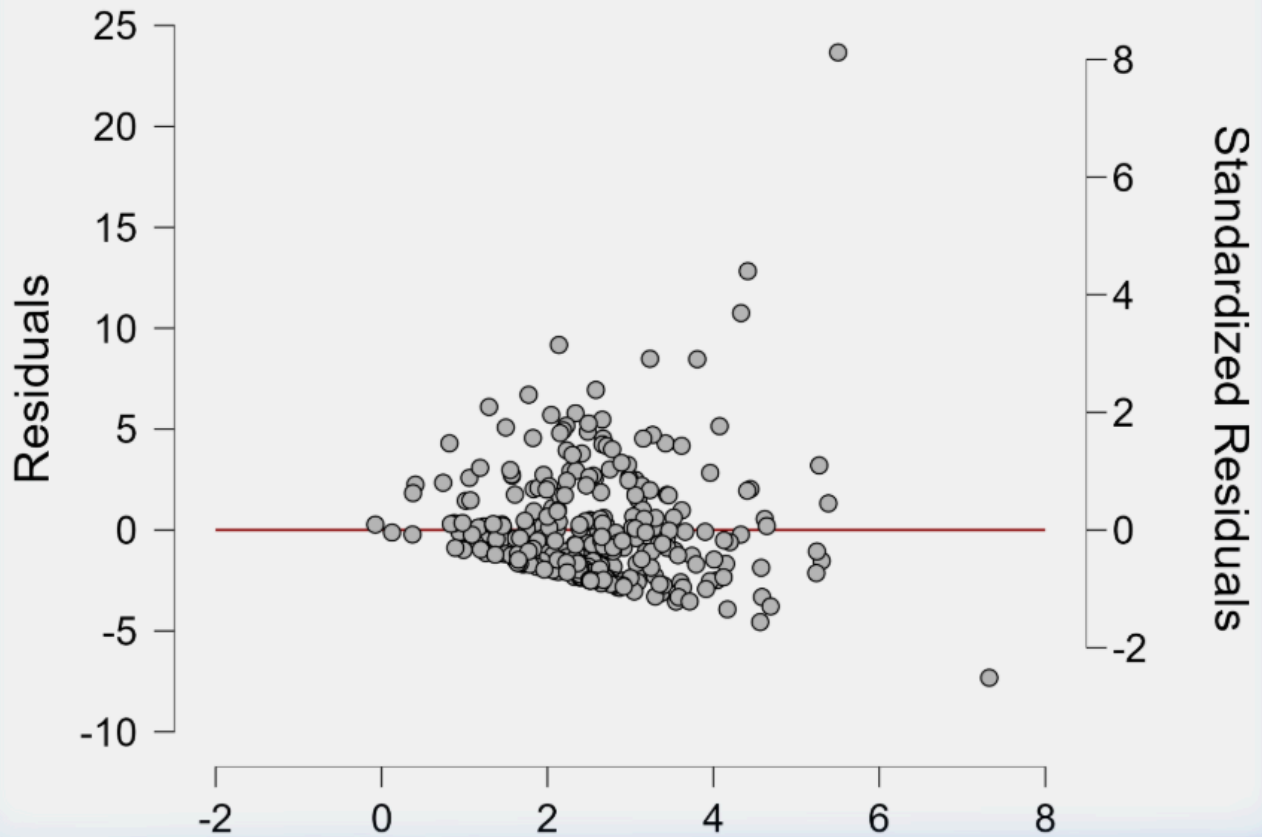


Figure 1: This plot demonstrates a clear "fan" pattern. As predicted suspension rates increase, the variance of the residuals also increases. This heteroscedasticity suggests that our model is less accurate at predicting outcomes in areas with high disciplinary rates and that our standard errors—and thus our p-values—may be biased.

Figure 2: Standardized Residuals Histogram (Non-Normality)

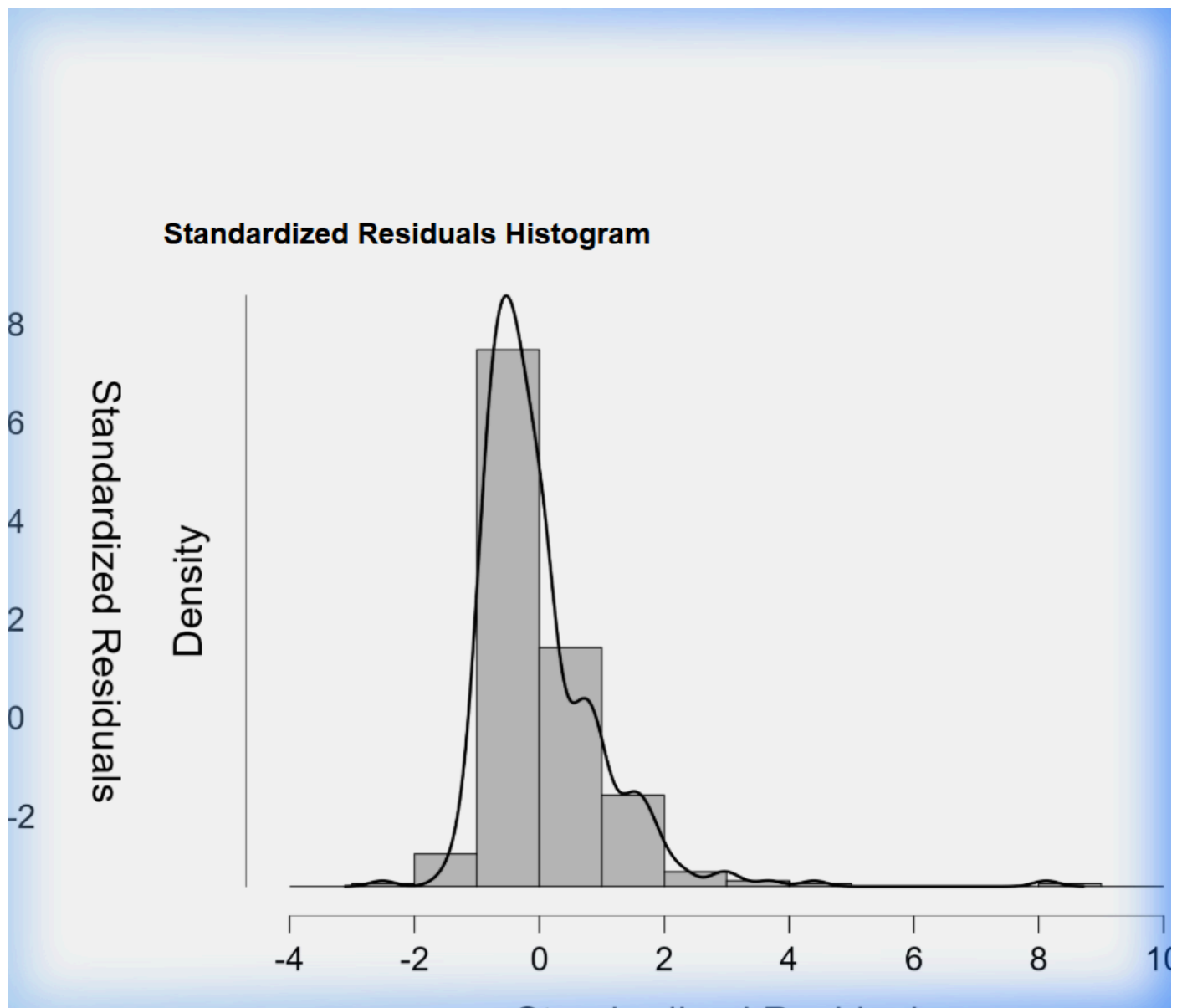


Figure 2: The histogram of standardized residuals is heavily right-skewed, showing a long tail of extreme positive outliers. This confirms that the disciplinary data does not follow a normal distribution, but likely follows a count-based distribution better suited for Generalized Linear Models (GLMs).

In conclusion, while our analysis tells a compelling story of how math proficiency and school attendance provide distinct pathways for reducing student suspensions, the violation of OLS assumptions suggests that these findings should be viewed as directional. Future research must adopt more sophisticated modeling techniques to fully capture the skewed and complex nature of school disciplinary data.